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Microstructure and properties of age-hardenable $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys

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ABSTRACT

Two promising high-entropy alloys (HEAs) $Al_xCrFe_{1.5}MnNi_{0.5}$ ($x = 0.3$ and 0.5) were designed from Al–Co–Cr–Cu–Fe–Ni alloys by substituting Mn for expensive Co and excluding Cu to avoid Cu segregation. Microstructures and properties were investigated and compared at different states: as-cast, as-homogenized, as-rolled and as-aged states. $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy in the as-cast, as-homogenized and as-rolled states has a dual-phase structure of BCC phase and FCC phase, in which Al, Ni-rich precipitates of B2-type BCC structure disperse in the BCC phase. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy in the corresponding states has a matrix of BCC phase in which Cr-rich particles of BCC structure and Al, Ni-rich precipitates of B2-type BCC structure disperse. These three BCC phases have the same lattice constant.

Both alloys are workable and show a hardness range of Hv 300–500 in the as-cast, as-forged, as-homogenized and as-rolled states. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy has a higher hardness level than $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ one because of its full BCC phase. Both alloys thus can be used as structural parts requiring stronger strength.

Both alloys display a significant high-temperature age-hardening phenomenon. As-cast $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy can attain the highest hardness, Hv 850, at 600 °C for 100 h, and $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ can get even higher hardness, Hv 890. The aging hardening is resulted from the formation of ρ phase ($Cr_5Fe_6Mn_8$-like phase). Prior rolling on the alloys before aging could significantly enhance the age-hardening rate and hardness level due to introduced defects. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy exhibits excellent oxidation resistance up to 800 °C, which is better than $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy. Combining this merit with its high softening resistance and wear resistance as compared to commercial alloys $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy has the potential for high-temperature structural applications.

1. Introduction

Almost all conventional alloys have been designed to have a major element, such as ferrous, aluminum, copper, titanium, and magnesium alloys. To extend this conventional concept of alloy designing, a new alloy field of multicomponent compositions was proposed in 1996, which has been coined as “high-entropy alloys (HEAs)”. A high-entropy alloy has been suggested to contain at least five principal metal elements, each of which has the content more than 5 at% but less than 35 at% [1,2]. Such alloys are featured with significantly higher mixing entropy than conventional alloys as compared at the liquid solution or regular solid solution state.

Based on the understanding and experience in metallurgy, an alloy with multi-principal elements was thought to be complicated in microstructure and difficult to be analyzed because many intermetallic compounds would form. They would be also brittle to be applied as structural materials [2,3]. However, with suitable alloy design, high-entropy alloys might exhibit unique features, such as (1) forming simple solid solution phases with nano- or even amorphous structure, (2) good thermal stability, (3) high hardness, and (4) superior corrosion and oxidation resistance [1,2,4–26]. They are expected to be abundant in academic research and industrial applications. Four core effects in high-entropy alloys would be more pronounced than those in conventional alloys [27]: (1) high entropy – it enhances the formation of simple solid solution phases, such as FCC and/or BCC structures, and thus simplifies the microstructures; (2) severe lattice distortion – it arises from the atomic size difference among different elements in solid solution phases and might markedly influence mechanical, physical and chemical properties; (3) sluggish diffusion – it is due to the inefficient cooperative diffusion of various species and thus slow down phase transformations; and (4) cocktail – it comes from the ideal mixing and inevitably excess interactions among the multi-principal elements in solid solution phases, and thus is a composite effect on properties.

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Table 1
Different processes applied for $Al_xCrFe_{1.5}MnNi_{0.5}$ HEAs.

States	Processes
As-cast	Arc melting and casting
As-forged	Arc melting and casting → forging at 1200 °C with a thickness reduction 50%
As-homogenized	Arc melting and casting → forging at 1200 °C with a thickness reduction 50% → homogenizing at 1100 °C for 6 h followed by air cooling
As-rolled	Arc melting and casting → forging at 1200 °C with a thickness reduction 50% → homogenizing at 1100 °C for 6 h followed by air cooling → warm rolling at 300 °C with 10%, 30%, 50%, and 80% reductions, respectively
As-aged	As-cast or as-rolled samples → aging at 600 °C, 700 °C, 800 °C, and 900 °C for 2 h, 5 h, 10 h, 20 h, 50 h, and 100 h, respectively

Previous studies on the $Al_{0.5}CoCrCuFeNi$ alloy demonstrated that it was of simple FCC phase and exhibits good hot and cold workability [9,10,20,24,26]. It displays high strength and high work-hardening capability up to 800 °C. However, its initial strength level is quite low, as an example, $Al_{0.5}CoCrCuFeNi$, has the as-cast hardness about 200 Hv which is similar to that of 304 and 316 stainless steels. Moreover, cobalt is an expensive and strategic material which might limit their broad applications.

This study is intended to develop a cheaper alloy system having higher strength and resistance to annealing softening. It was designed to be $Al_xCrFe_{1.5}MnNi_{0.5}$ (x = 0.3 and 0.5) alloy based on Al–Co–Cr–Cu–Fe–Ni alloys by substituting Co with Mn referring to the fact that Mn brings good merits for austenite manganese steels [28], and excluding Cu to avoid the Cu segregation in the interdendrite regions [9,10]. In addition, the Fe content is increased so as to further reduce the cost. Two Al contents were chosen for this study because they exhibit pronounced high-temperature aging hardening effect as compared with those Al contents lower than 0.3 or higher than 0.5. These two alloys were investigated on various states: as-cast, as-forged, as-homogenized, as-rolled and as-aged states. The results show that the new alloy system has its uniqueness and the potential in applications.

2. Experimental procedure

The $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys (x = 0.3 and 0.5) were prepared by arc melting elemental Al, Cr, Fe, Mn, and Ni raw materials at a current of 500 A in a water-cooled copper hearth. The source materials had a purity level higher than 99.0 wt%. Melting was performed at a pressure of 0.01 atm after purged with argon three times. Repeated melting for at least four times was done to improve chemical homogeneity of the alloys. The ingots were approximately 30 mm in diameter and 10 mm in thickness and 50 g in weight. First, they were forged with a thickness reduction of 50% at 1200 °C and homogenized at 1100 °C for 6 h followed by air cooling. Then, they were warm rolled at 300 °C with 10%, 30%, 50%, and 80% reductions, respectively. The as-cast and as-rolled samples were annealed at 600 °C, 700 °C, 800 °C, and 900 °C for 2 h, 5 h, 10 h, 20 h, 50 h, and 100 h, respectively. Table 1 summarizes the details of different processes.

A JEOL-5410 scanning electron microscope (SEM) was used to analyze the microstructure and compositions. X-ray mapping was done by high resolution hyper electron-probe microanalysis (FE-EPMA, JEOL JXA-8500F, Tokyo, Japan) to gain the distribution of elements of test sample. X-ray diffraction (XRD) patterns were obtained by a Rigaku ME 510-FM2 diffractometer operated at 30 kV and 20 mA. The range of scanning angle (2θ) was from 20° to 100°

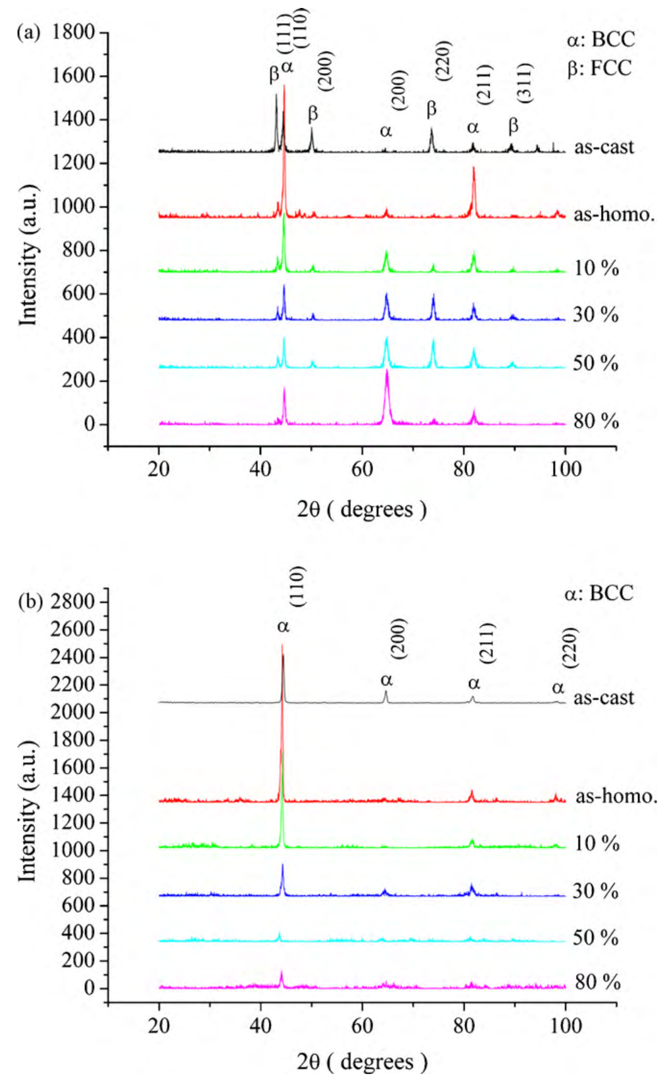

Fig. 1. XRD analyses of $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys in the as-cast, as-homogenized, and as-rolled states: (a) x = 0.3 and (b) x = 0.5.

and the scanning rate was 4°/min. Vickers hardness measurement was conducted with a Matsuzawa Seiki MV-1 tester under a loading of 5 kg and a duration time of 15 s. Seven measurements were made on each sample for an average. Hot Vickers hardness measurement was performed with an Akashi AVK-HF hardness tester at RT, 300 °C, 500 °C, 700 °C, and 800 °C, respectively.

Oxidation resistance was obtained by isothermal oxidation treatment at 800 °C for different time intervals. Each datum point of weight gain was obtained by weighing the oxidized specimens after the treatment. Wear resistance was measured by the pin-on-belt wear test [4,6]. The belt was made of the Al_2O_3 emery cloth with grit #100. The testing was performed at a speed of 50 cm/s, a wear distance of 20 m, and a load of 5 kg. The wear resistance was calculated using the formula: 1/[(the mass of loss/density) × 1000/wear distance], and has a unit of m/mm^3.

3. Results and discussion

3.1. X-ray diffraction analysis and hardness comparison

Fig. 1 shows the XRD patterns of $Al_xCrFe_{1.5}MnNi_{0.5}$ alloy in the as-cast, as-homogenized, and as-rolled states, respectively. The crystal structures of the $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloys in all states are

Table 2
Vickers hardness values (Hv) of $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys in different states.

Alloys	As-cast	As-forged	As-homo.	As-rolled 10%	As-rolled 30%	As-rolled 50%	As-rolled 80%
$Al_{0.3}CrFe_{1.5}MnNi_{0.5}$	297 ± 2	358 ± 6	339 ± 16	359 ± 13	386 ± 7	407 ± 8	446 ± 8
$Al_{0.5}CrFe_{1.5}MnNi_{0.5}$	396 ± 7	428 ± 6	394 ± 10	428 ± 8	457 ± 11	464 ± 14	480 ± 6

of FCC and BCC as shown in Fig. 1(a). The lattice constants of FCC and BCC are 3.63 Å and 2.88 Å, respectively. The volume fraction of FCC in the as-homogenized state is much less than that in the cast state as reflected by the relative peak intensity of FCC to that of BCC. This indicates that there are excess amount of FCC phase formed during the nonequilibrium solidification. Rolling deformation caused the peak intensity to decrease. This is attributable to the increased diffusion scattering due to the increased lattice distortion caused by deformation. Besides, it is noted that the (2 0 0) peak intensity of BCC phase increases relative to other peak intensities. This indicates the increase in the degree of (2 0 0) texture, *i.e.* more (2 0 0) planes are aligned in parallel with the rolling plane. From Fig. 1(b), it can be seen that the crystal structures of $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloys in all states are of BCC. The lattice constant is 2.88 Å. This increase of BCC phase for the higher Al content agrees with the fact that aluminum is a BCC former due to its stronger p–d orbital bonding with transition metals [9]. Diffusion scattering enhanced by increasing rolling deformation is also found.

Table 2 lists the hardness of $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys in the as-cast, as-forged, as-homogenized and as-rolled states. Higher Al content gives a higher hardness due to its stronger p–d orbital bonding with transition metals and the formation of more BCC phase. Both alloys show small work hardening after hot forging. A homogenization treatment at 1100 °C for 6 h slightly lowers the hardness. Rolling deformation at 300 °C increases the hardness reflecting a moderate work hardening in warm rolling. Both alloys are thus workable and show a hardness range of Hv 300–500 in the as-cast, as-forged, as-homogenized and as-rolled states and can be used as structural parts requiring stronger strength.

3.2. High-temperature age-hardening phenomenon

Both alloys exhibit high-temperature aging hardening. The hardness variations of the as-cast samples with aging time at 600 °C, 700 °C, 800 °C, and 900 °C are shown in Fig. 2. $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy exhibits a large age-hardening effect at temperatures from 600 °C to 800 °C but a smaller age-hardening at 900 °C. The hardness increases from Hv 300 to Hv 850 in 100 h at 600 °C. It takes about 5 h to attain the peak values: Hv 800 for 700 °C and Hv 735 for 800 °C. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy also exhibits the high-temperature age-hardening phenomenon. But the hardening is strong at 600 °C and 700 °C, moderate at 800 °C, negligible at 900 °C. For 600 °C, the hardness rises to about Hv 890 at 100 h. For 700 °C, the hardness reaches the peak about Hv 840 at 50 h. For 800 °C, the hardness reaches the peak Hv 575 at 50 h. The cause for the aging hardening phenomenon can be revealed by Fig. 3 which compares the XRD patterns between the as-cast sample and the as-aged sample at 700 °C for 20 h. The multiple peaks existing between 40° and 50° are identified to be from the crystal structure of $Cr_5Fe_6Mn_8$ (ρ phase) according to the JCPDS card. The ρ phase has a tetragonal structure with lattice constants of $a = 9.09$ Å and $c = 9.99$ Å. As the formation of ρ phase is accompanied with a decreased amount of BCC phase, ρ phase is thus transformed from the BCC phase. In order to see the role of ρ phase in hardening, mono ρ phase with the composition $Cr_5Fe_6Mn_8$ has been prepared by vacuum arc melting and found to display a very high hardness of Hv 1273. Thus, we attribute the age-hardening to the formation of ρ phase transformed from BCC phase. This strong high-temperature aging hardening phenomenon is seldom found in conventional alloys. Similar phenomenon was reported for

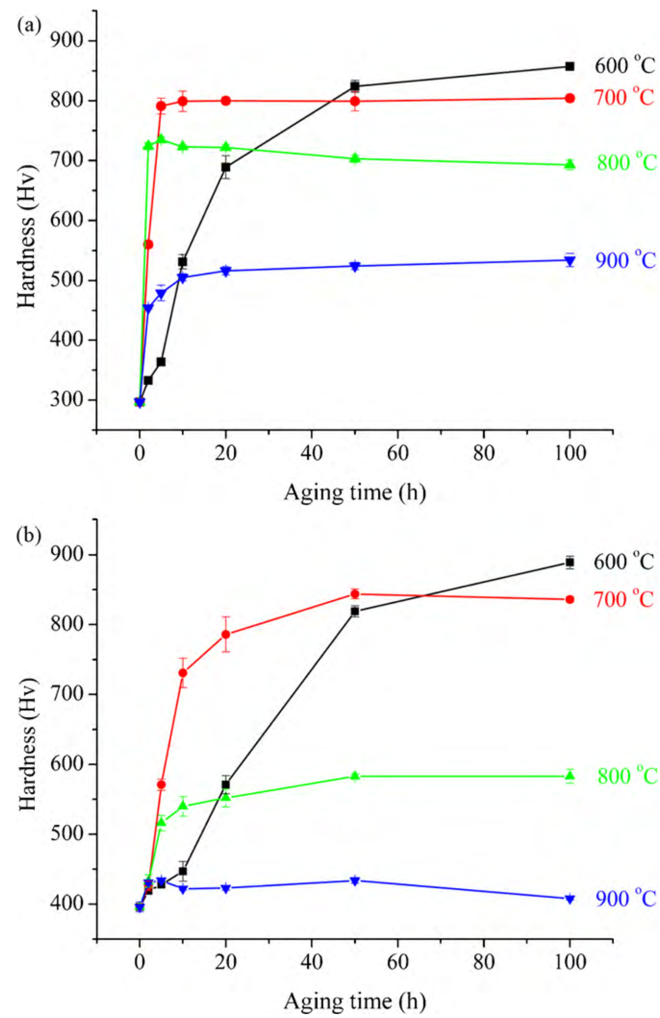

Fig. 2. Aging hardness curves of as-cast $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys aged at 600 °C, 700 °C, 800 °C, and 900 °C, respectively: (a) $x = 0.3$ and (b) $x = 0.5$.

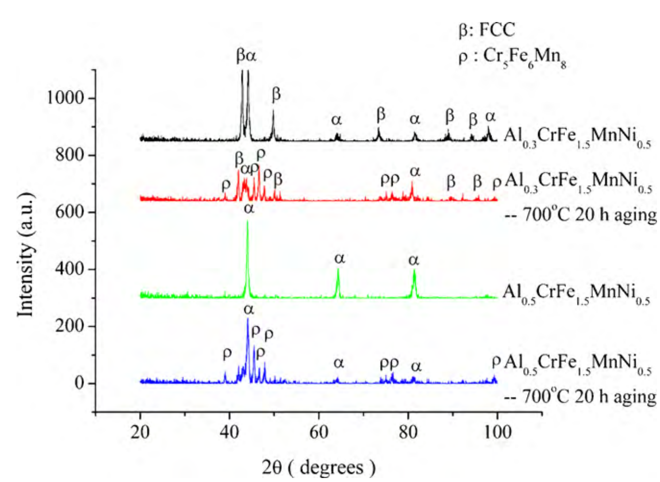

Fig. 3. XRD patterns of the as-cast sample and the sample aged at 700 °C for 20 h.

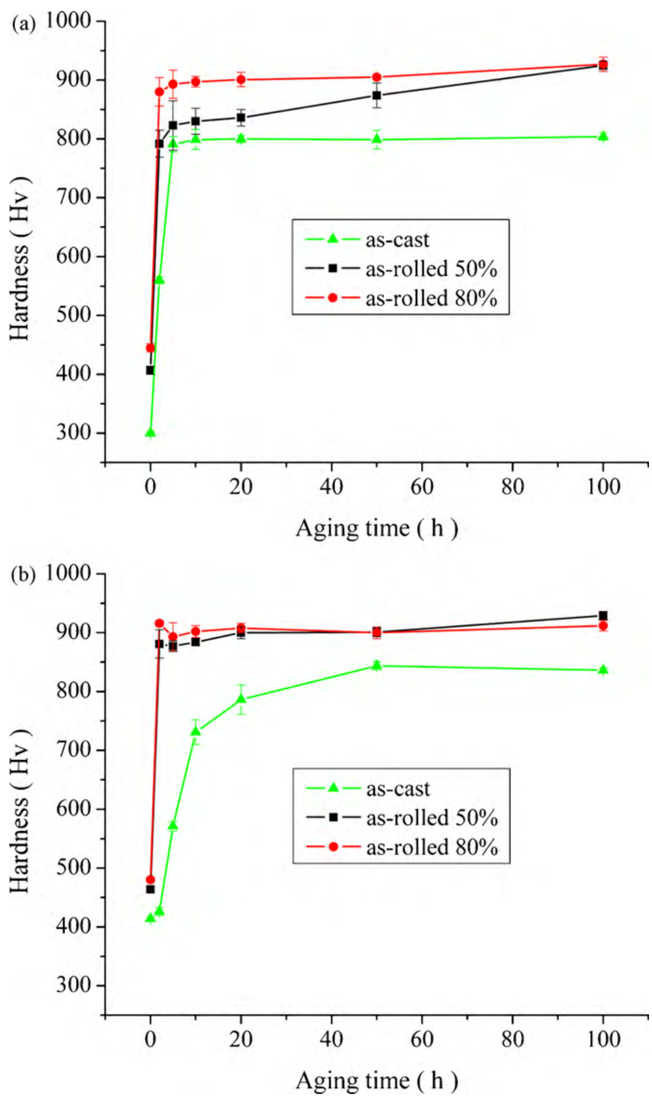

Fig. 4. Aging hardening curves obtained by aging as-rolled $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys at 700 °C: (a) x = 0.3 and (b) x = 0.5.

the thermal sprayed coatings of two kinds of high-entropy alloys [4].

Fig. 4 shows the aging hardening curves of as-rolled $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys aged at 700 °C. It can be seen that the age-hardening rate and peak hardness level can be enhanced by rolling deformation. The peak hardness reaches the same level

Table 3
Chemical compositions of different regions in as-cast $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys.

Alloys	Elements (at.%)	Al	Cr	Fe	Mn	Ni
$Al_{0.3}CrFe_{1.5}MnNi_{0.5}$	Dendrite	8.1	24.6	35.1	21.3	10.9
	Interdendrite	6.3	22.8	35.6	23.2	12.1
$Al_{0.5}CrFe_{1.5}MnNi_{0.5}$	BCC matrix	12.5	19.2	33.5	23.7	11.1
	Cr-rich phase	7.8	44.8	20.7	16.5	10.2

around Hv 900 for both alloys either with 50% or 80% rolling reduction. The increase in age-hardening rate and peak hardness level are attributed to considerable quantities of defects, *e.g.* vacancies and dislocations introduced by rolling deformation, which provide more heterogeneous nucleation sites of ρ phase, enhance the diffuse rate, and thus accelerate the formation of ρ phase.

3.3. Microstructure analysis

Fig. 5 shows the SEM microstructures of as-cast $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys. Table 3 lists the chemical compositions of different regions. The as-cast structure $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy consists of dendrite and interdendrite, shown in Fig. 5(a). Dendrite region has higher Al and Cr contents than interdendrite region. Ke et al. [29] studied the Al–Co–Cr–Cu–Fe–Ni high-entropy alloy system and found that aluminum and chromium are BCC stabilizers whereas nickel and cobalt are FCC stabilizers, as found in ferrous alloys. Therefore, the dendrite region is BCC phase whereas interdendrite is FCC phase. A close-up view shows a distribution of fine gray precipitates in the dendritic matrix, which will be identified to be rich in Al and Ni later. The precipitates are thus a solution phase of the NiAl compound (B2-type BCC with the lattice constant same as that of the BCC matrix phase) in which there have elemental substitutions, *i.e.* Fe and Mn for partial Ni, and Cr for partial Al. This phenomenon was also reported in the $Al_xCoCrCuFeNi$ alloys [9].

The as-cast microstructure $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy consists of main dendrite region and minor interdendrite region which is a eutectic structure. According to Table 3, large particles in the eutectic have Cr content of 44.8 at% much higher than the dendrite region. Based on the X-ray analysis, all these solid solution phases, dendrite phase (or matrix phase), Cr-rich phase, and Al, Ni-rich fine gray precipitates in the dendrite phase, are of BCC phase having the same lattice constant but different compositions.

In order to get coarser precipitates dispersed in the dendrite matrix for identifying their chemical composition, cast $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy was homogenized at 1100 °C for 2 h

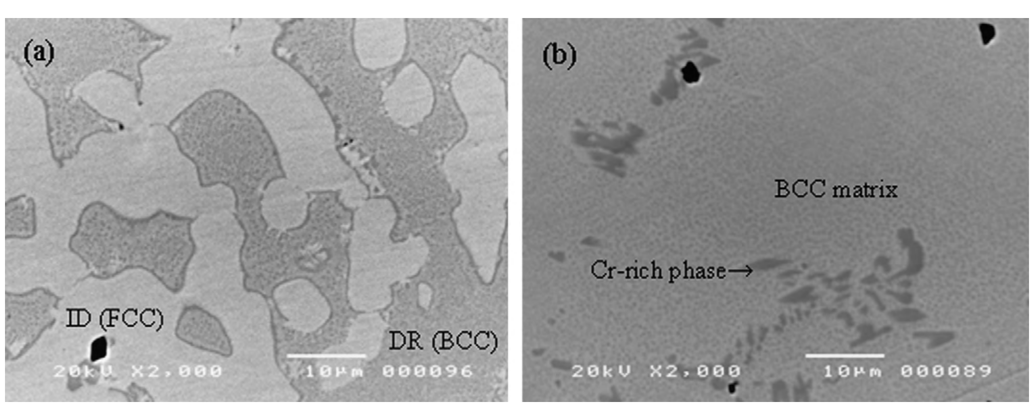

Fig. 5. SEM microstructures of as-cast $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys: (a) x = 0.3 and (b) x = 0.5 (DR: dendrite, ID: interdendrite).

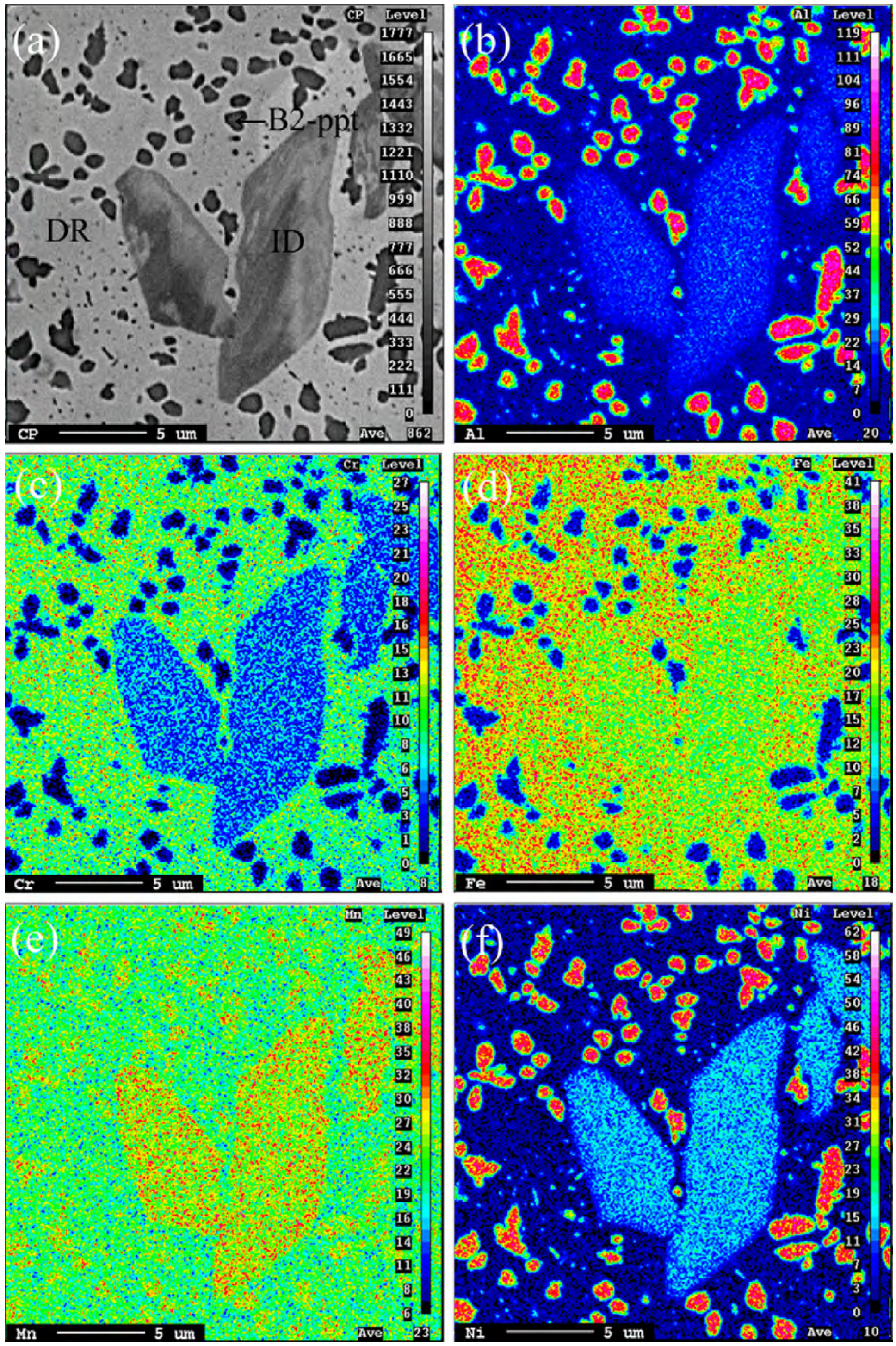

Fig. 6. FE-EPMA X-ray mappings of cast $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy homogenized at 1100 °C for 2 h followed by furnace cooling: (a) BEI image and (b) through (f) elemental color mapping for Al, Cr, Fe, Mn, and Ni, respectively (DR: dendritic matrix, ID: interdendrite).

followed by furnace cooling. Elemental color mapping analyzed by FE-EPMA was used to reveal the distribution of elements in $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy. Fig. 6 shows elemental color mappings of cast $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy which have received homogenization and furnace cooling. It clearly reveals that the micro-precipitates with a size 1–2 μm dispersed in the dendrite matrix are Al, Ni-rich phase. It is also noted that the dendrite matrix is rich in Cr, Mn, and Fe and the interdendrite contains more Fe, Mn and Ni than overall dendritic phase, which agrees with the results shown in Table 3.

Figs. 7 and 8 compare the OM microstructures on the long-transverse cross sections of $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ and $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloys with different rolling reductions. For the as-homogenized $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy, the grains become thinner under larger rolling deformation. The longish particles which correspond to the softer FCC interdendrite phase are distributed in the matrix and along grain boundaries, and align as stringers in parallel to the rolling direction with 80% reduction. As for the $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy, the Cr-rich particles

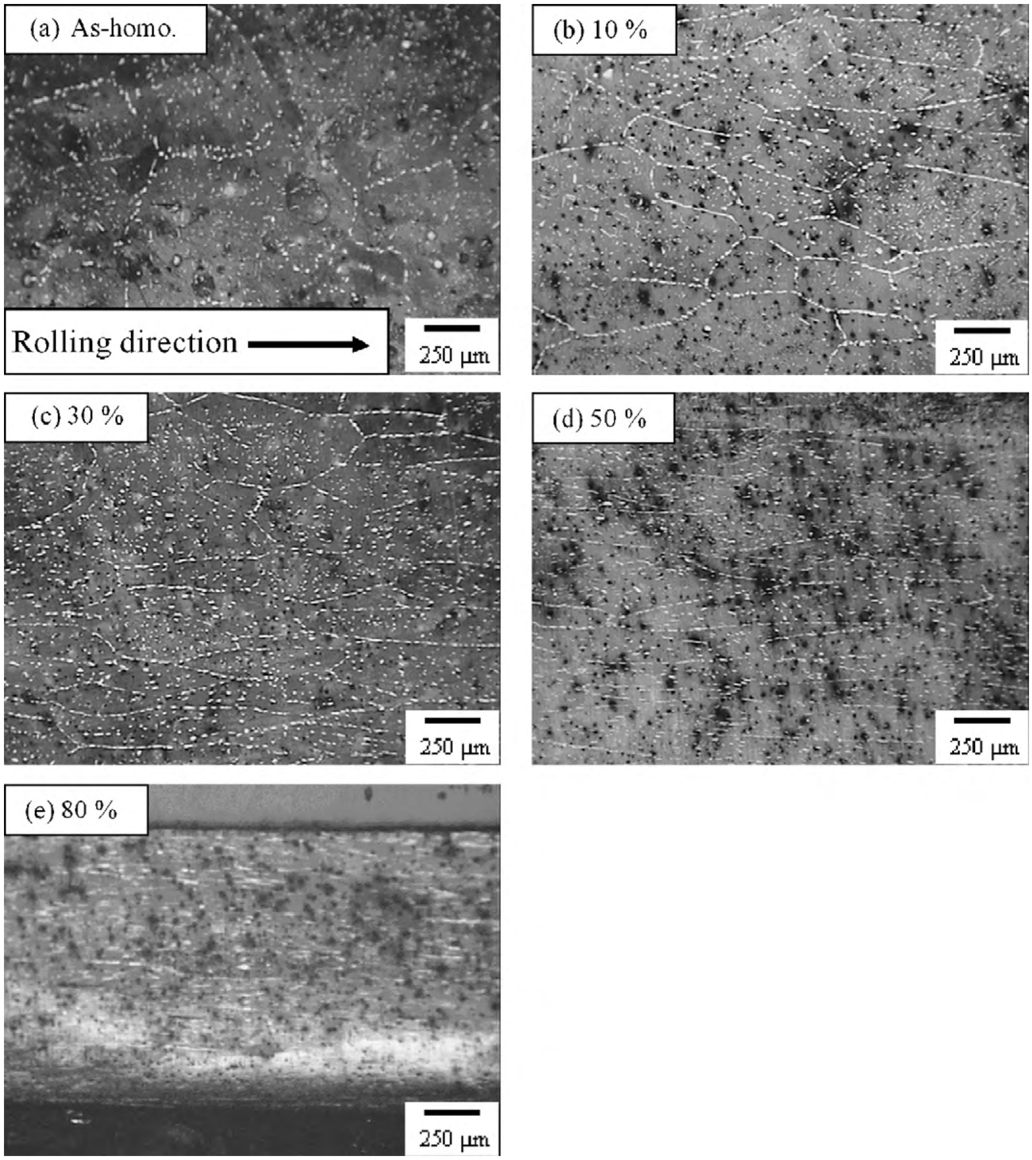

Fig. 7. OM images of long-transverse sections of the $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy samples rolled with different reductions.

(white spots) initially distributed along grain boundaries and in the matrix are elongated toward the rolling direction for smaller rolling reductions but broken under 50% deformation and spread spatially under 80% deformation. Based on the observation of as-forged and as-rolled samples, it reveals that both alloys possess good workability.

3.4. Properties for high-temperature applications

3.4.1. Oxidation resistance

Fig. 9 is the weight gain curve of $Al_xCrFe_{1.5}MnNi_{0.5}$ alloy oxidized at 800 °C. It is interesting to see the initial oxidation stage follows the parabolic curve indicating the oxidation is diffusion-controlled. Since the latter stage approach a limit, the oxidation turns into the interface-controlled process. This is expected since the EDS analysis shows the outmost layer is of Mn oxide and the undersurface is of dense and protective Cr and Al oxides. Moreover, it can be seen that $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy has a superior oxidation resistance to $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ one due to its higher aluminum content. This result demonstrates that $Al_xCrFe_{1.5}MnNi_{0.5}$ alloy has a good oxidation resistance at 800 °C, however the oxidation resistance of $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy is better than that of $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ one.

3.4.2. Hot hardness comparison

For the assessment on the elevated-temperature applications, hot hardness of $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy with better oxidation resistance was measured. Fig. 10 compares the $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy with high-carbon steels and high-speed steels [30]. To get higher room-temperature hardness the present alloy was preaged at 700 °C for 10 h. The hardness decreases from Hv 740 at RT to Hv 430 at 700 °C and Hv 340 at 800 °C. Comparatively, the hardness of high-carbon steels decreases significantly when the temperature is more than 300 °C and that of high-speed steels is down to Hv 250 at 700 °C. Thus, the present alloy is better in hot hardness and has potential in high-temperature molds and tools.

3.4.3. Wear resistance comparison

Fig. 11 compares the combination of wear resistance and hardness for $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloys in the as-homogenized and as-aged states, and commercial alloys: cast irons, 316 and 17-4PH

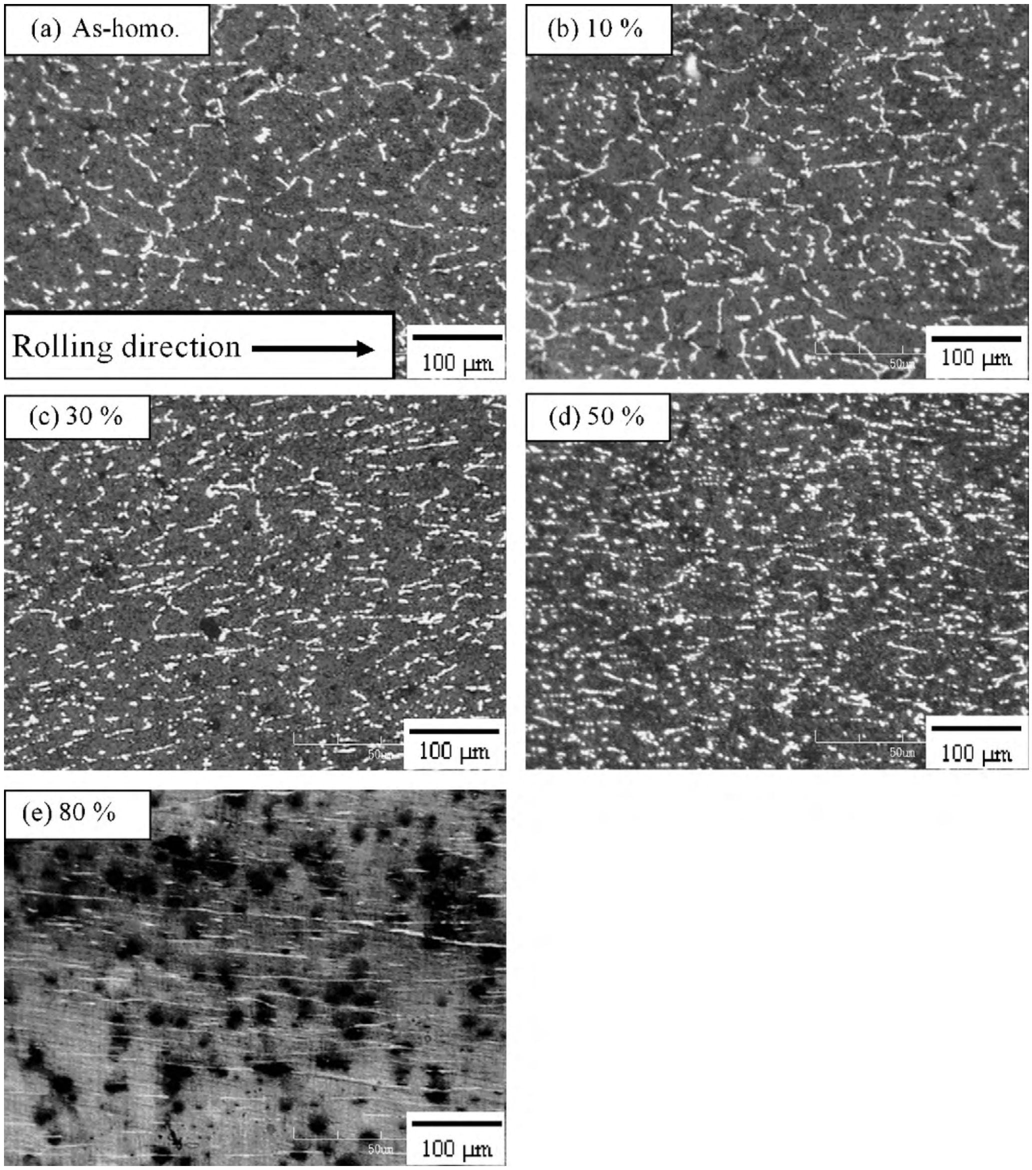

Fig. 8. OM images of long-transverse sections of the $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy samples rolled with different reductions.

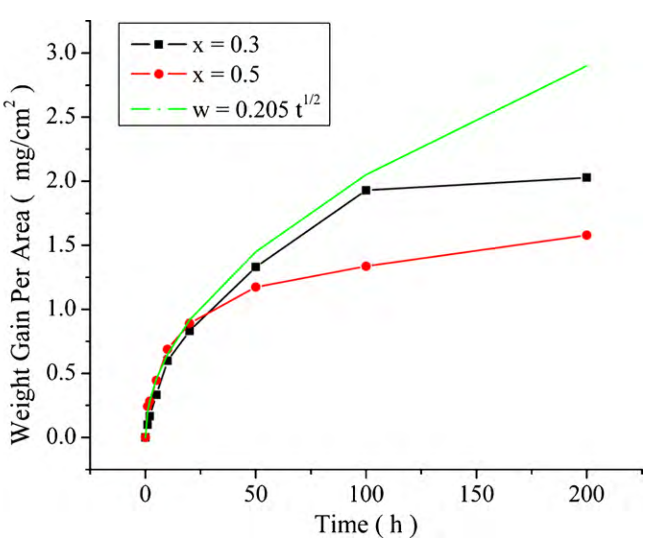

Fig. 9. Weight gain curves of the $Al_xCrFe_{1.5}MnNi_{0.5}$ alloys oxidized at 800 °C.

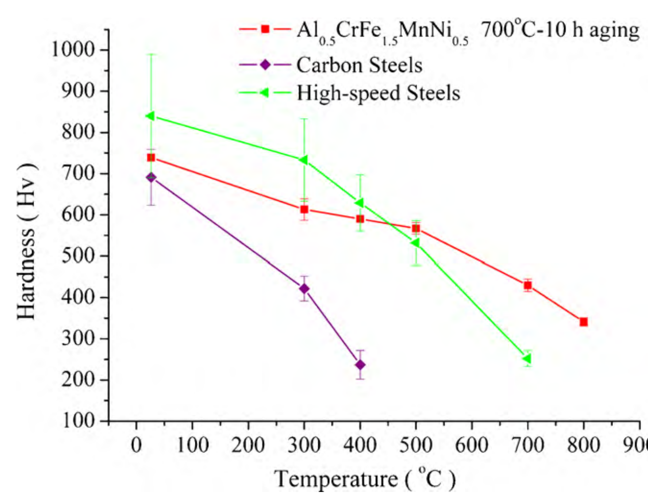

Fig. 10. Hot hardness of the $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy aged at 700 °C for 10 h, high-carbon steels, and high-speed steels.

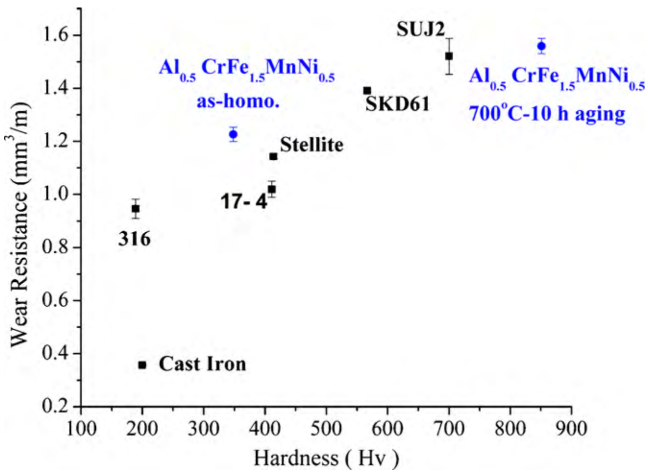

Fig. 11. Comparison of wear resistance and hardness between $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloys in the as-homogenized and as-aged states, and several commercial alloys.

stainless steels, Stellite 6, SKD61 tool steels, and SUJ2 bearing steels. It reveals that the present alloy in the as-homogenized state has smaller hardness but higher wear resistance than that of 316 and 17-4PH steels and Stellite 6. The alloy in the as-aged state has a higher hardness but a similar wear resistance level to tool steels and bearing steels.

4. Conclusions

1. $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ alloy in the as-cast, as-homogenized and as-rolled states has a dual-phase structure of BCC phase and FCC phase, in which Al, Ni-rich precipitates of B2-type BCC structure disperse in the BCC phase. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy in all states has a matrix of BCC phase in which Cr-rich particles of BCC structure and Al, Ni-rich precipitates of B2-type BCC structure disperse. These three BCC phases have the same lattice constant.
2. $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy has a higher hardness level than $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ one in the as-cast, as-forged, as-homogenized and as-rolled states. This is because the former alloy fully consists of BCC phase and does not contain FCC phase. Both alloys in the as-homogenized state can be warm rolled up to 80% thickness reduction and display moderate work hardening.
3. Both alloys display a significant high-temperature age-hardening phenomenon which is seldom found in conventional alloys. The as-cast samples can attain about Hv 850 by aging at 600–800 °C. The hardening comes from the formation of hard ρ phase.
4. Rolling deformation can accelerate age-hardening rate and increase peak hardness as compared with the as-cast samples. This is attributed to considerable quantities of defects, *e.g.* vacancies and dislocations introduced by rolling deformation, which provide more heterogeneous nucleation sites of ρ phase, enhance the diffuse rate, and thus accelerate the formation of ρ phase.
5. Both alloys have good oxidation resistance up to 800 °C, however $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy is better than that of $Al_{0.3}CrFe_{1.5}MnNi_{0.5}$ one due to its higher Al content.
6. Age-hardened $Al_{0.5}CrFe_{1.5}MnNi_{0.5}$ alloy exhibits higher softening resistance than high-speed steels, and good wear resistance as compared with commercial alloys, it has the potential in high-temperature structural applications.

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